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Systemic Autonomic and Cytostructural Dysregulation in the Human Body Under Vibration: A Frequency-Dependent Analysis

1. Introduction: The Bio-Vibrational Interface

The interaction between mechanical vibration and the human biological system represents a sophisticated interface of physics and physiology, where kinetic energy is not merely absorbed as trauma but transduced into complex neurovascular signals. For decades, the study of Hand-Arm Vibration Syndrome (HAVS) and Vibration-Induced White Finger (VWF) was dominated by a localized injury model, presuming that the damage observed in the fingers of industrial workers—chain saw operators, grinders, and rock drillers—was the direct result of mechanical battering of the digital tissues. However, a pivotal shift in understanding emerged from the research presented in the *Nagoya Journal of Medical Science* (Volume 57, Supplement, 1994) and its subsequent lineage of investigation. This body of work systematically dismantled the exclusively local theory, replacing it with a nuanced "somato-sympathetic" model. This model posits that local vibration, particularly at specific frequencies such as 60 Hz, acts as a potent trigger for central autonomic reflexes, leading to systemic sympathetic dysregulation that manifests not only in the exposed hands but also in the unexposed feet.¹

The implications of this frequency-dependent interaction extend far beyond occupational health, touching upon fundamental principles of human neurophysiology, mechanotransduction, and vascular remodeling. This report provides an exhaustive, expert-level synthesis of these interactions. It integrates microneurographic data, histological evidence of arterial hypertrophy, and biodynamic resonance characteristics to construct a comprehensive picture of how the human body processes and reacts to oscillatory mechanical forces. By examining the discrepancy between biological responses and established safety standards (ISO 5349), this analysis also highlights the critical need for a frequency-weighted approach that aligns with the physiological realities of the Pacinian and Meissner mechanoreceptor channels.⁴

1.1 Historical Trajectory and Clinical Definition

The recognition of vibration as a pathogen dates back to the early 20th century, with reports

of "dead finger" in workers using pneumatic tools. In Japan, research began in the 1930s, focusing on the railway, mining, and shipyard sectors.¹ The condition encompasses a triad of disorders: vascular (Raynaud's phenomenon), neurological (numbness and reduced sensation), and musculoskeletal (weakness and pain). The *Nagoya International Symposium on Human Responses to Hand-Arm Vibration* (1993) marked a watershed moment, consolidating evidence that these symptoms were manifestations of a "complex autonomic dysfunction" rather than simple tissue necrosis.¹ The symposium's proceedings highlighted that the sympathetic nervous system is not a monolith but possesses distinct, frequency-tuned pathways for skin and muscle regulation, a finding that remains central to understanding the etiology of vibration syndrome.¹

2. Biomechanics of the Hand-Arm System: Impedance, Resonance, and Transmission

Before analyzing the physiological cascade, one must understand the physical entry point: the coupling of the vibrating tool to the human hand. The hand-arm system is a complex mechanical structure consisting of masses (tissue segments), springs (ligaments and skin elasticity), and dampers (viscoelastic soft tissues). When vibration is applied, the body's response is governed by its mechanical impedance and transmissibility, which are highly frequency-dependent.

2.1 Mechanical Impedance and Energy Absorption

Mechanical impedance describes the resistance of the hand-arm system to imposed motion. It is defined as the ratio of the driving force to the resulting velocity at the point of input. Research indicates that the hand-arm system behaves non-linearly. At low frequencies (<50 Hz), the system acts relatively stiffly, transmitting vibration efficiently past the wrist to the elbow and shoulder. This transmission is implicated in the musculoskeletal complaints of the proximal joints often seen in heavy tool operators.⁶

However, at higher frequencies, the impedance characteristics change. The soft tissues of the palm and fingers begin to decouple from the skeletal structure, acting as isolators. While this protects the shoulder, it means that the high-frequency energy is absorbed almost entirely within the hand and fingers. This localized absorption is critical because energy absorbed is energy available to cause damage. The "biodynamic hypothesis" suggests that the risk of injury is proportional to the vibration power absorbed per unit volume of tissue. Consequently, high-frequency vibration, while less transmissible to the arm, delivers a high energy density to the digital vessels and nerve endings.⁸

2.2 Resonance Frequencies and Amplification

Resonance occurs when the frequency of the external force matches the natural frequency of the body segment, leading to amplification of the oscillation. This phenomenon is central to

the "frequency-dependence" of injury.

- **Whole Hand-Arm Resonance:** The fundamental resonance of the hand-arm system (operating as a single unit relative to the handle) is typically identified in the range of **16 Hz to 62.5 Hz**.⁷ Finite element modeling and experimental modal analysis confirm that significant strain is developed in the wrist and elbow structures within this band.⁷ This frequency range overlaps precisely with the 60 Hz band identified in microneurographic studies as the most potent trigger for sympathetic reflexes¹, creating a "double jeopardy" where mechanical amplification and neurological sensitivity coincide.
- **Digital Resonance:** The fingers, having lower mass and high stiffness relative to the arm, exhibit resonance at much higher frequencies. Studies using laser vibrometry have mapped the resonant modes of individual fingers to the **100 Hz to 300 Hz** range.⁶ Specifically, the proximal and distal phalanges may vibrate out of phase with the handle at these frequencies, subjecting the digital arteries and nerves to intense shear and tensile stresses.
- **Axis-Specific Response:** The resonance is not uniform across all axes. For instance, in the Z-direction (dorsal-volar), resonance magnitudes are generally lower, while the Y-direction (transverse) can exhibit transmissibility ratios exceeding 3.0 at resonance.⁶ This anisotropic response complicates the assessment of risk, as the dominant axis of vibration varies by tool type.

2.3 The Damping Role of Soft Tissue

The glabrous skin of the palm and fingers is specialized for gripping and shock absorption. However, under continuous vibration, the viscoelastic properties of these tissues can be altered. High-frequency vibration (>100 Hz) essentially "liquefies" the response of the subcutaneous fat pads, leading to efficient propagation of shear waves along the skin surface.⁸ This surface transmission is particularly relevant for the activation of superficial mechanoreceptors like Meissner's corpuscles, which are embedded in the dermal papillae just below the epidermis.

3. Neurophysiology of Mechanoreception: The Peripheral Encode

The transduction of mechanical vibration into a neural code is the first physiological step in the vibration syndrome cascade. The glabrous skin of the human hand is innervated by four distinct classes of low-threshold mechanoreceptors (LTMRs), but the literature confirms that two types—Meissner's corpuscles and Pacinian corpuscles—are the primary transducers of vibration.

3.1 Receptor Tuning and Channel Separation

The somatosensory system separates vibratory stimuli into two distinct perceptual channels:

"flutter" and "vibration."

- **The Flutter Channel (Meissner's Corpuscles / FA-I):** Meissner's corpuscles are located superficially in the dermal papillae, densely populating the fingertips. They are innervated by Fast-Adapting Type I (FA-I) fibers. These receptors function as velocity detectors and are tuned to low-frequency oscillations. Their optimal sensitivity lies between **10 Hz and 60 Hz**.¹⁰ They are responsible for the sensation of flutter or skin slip. The Nagoya studies' finding that 60 Hz is the peak frequency for driving skin sympathetic nerve activity (SSNA) strongly implicates the FA-I channel as the primary afferent limb of the somato-sympathetic reflex.³ The 60 Hz stimulus represents the upper bound of the Meissner bandwidth, potentially maximizing the discharge rate before the receptor saturation point or handover to the Pacinian system.
- **The Vibration Channel (Pacinian Corpuscles / FA-II):** Pacinian corpuscles are large, onion-like structures located deep in the dermis and subcutaneous tissue. Innervated by Fast-Adapting Type II (FA-II) fibers, they are acceleration detectors with extreme sensitivity. They can detect skin indentations as small as 10 nanometers. Their frequency response range is broad, from **60 Hz to over 700 Hz**, with a peak sensitivity around **250 Hz**.¹³ While Meissner's corpuscles detect the "thud" of the tool, Pacinian corpuscles detect the "buzz" or "whine." High-frequency vibration from tools like grinders (often >6000 RPM, i.e., >100 Hz) primarily activates this channel.

3.2 Molecular Mechanotransduction: The Piezo2 Channel

Recent advances have elucidated the molecular machinery underlying this sensitivity. The **Piezo2** ion channel has been identified as the principal mechanotransduction channel in both Meissner and Pacinian corpuscles.¹⁵ Piezo2 is a mechanically gated cation channel that opens in response to tension in the cell membrane, allowing rapid depolarization.

Research indicates that Piezo2 activity is not only required for acute sensation but is also essential for the *development and maintenance* of the end-organs themselves. In the absence of Piezo2-mediated activity, Meissner and Pacinian corpuscles fail to mature structurally.¹⁵ This developmental plasticity suggests a potential mechanism for the sensorineural damage seen in HAVS. Chronic, high-intensity activation of Piezo2 channels by industrial vibration may lead to excitotoxic calcium influx, receptor desensitization, or structural regression of the corpuscle—a phenomenon termed "terminal exhaustion." This aligns with clinical findings of reduced vibrotactile perception thresholds in workers with VWF.¹⁶

Table 1: Comparative Characteristics of Vibration Mechanoreceptors

Feature	Meissner's Corpuscle (FA-I)	Pacinian Corpuscle (FA-II)

Location	Superficial Dermis (Dermal Papillae)	Deep Dermis / Subcutaneous Tissue
Receptive Field	Small, Sharp Borders	Large, Diffuse Borders
Optimal Frequency	10 – 60 Hz (Flutter)	60 – 700 Hz (Vibration)
Peak Sensitivity	~30-40 Hz	~250 Hz
Threshold	Moderate Indentation (<10 μm)	Extremely Low Indentation (<1 μm)
Role in HAVS	Trigger for SSNA Reflex (60 Hz)	Primary site of High-Freq Damage
Molecular Channel	Piezo2	Piezo2

4. The Autonomic Interface: Microneurography and the Somato-Sympathetic Reflex

The most technically sophisticated insights from the Nagoya studies involve the use of microneurography to dissect the sympathetic nervous system's response to vibration. This technique allows for the direct recording of muscle sympathetic nerve activity (MSNA) and skin sympathetic nerve activity (SSNA) from peripheral nerves (e.g., the tibial or peroneal nerve) in awake humans.

4.1 The Functional Dichotomy of SSNA and MSNA

Historically, the sympathetic nervous system was viewed as a "mass discharge" system—the fight-or-flight response. However, microneurography revealed that sympathetic outflows to different tissues are independently regulated.

- MSNA (Muscle Sympathetic Nerve Activity):** MSNA is primarily under baroreflex control. Its bursts are pulse-synchronous, driven by the need to maintain arterial blood pressure. It responds to gravity (+Gz stress), hypobaric hypoxia, and static muscle exertion (via metaboreflexes).¹ Crucially, the Nagoya researchers demonstrated that **MSNA is NOT enhanced by local vibration of the body.**¹ This is a vital negative finding. It indicates that vibration does not trigger a generalized, hemodynamic vasoconstriction (which would raise systemic blood pressure) but rather a selective response.
- SSNA (Skin Sympathetic Nerve Activity):** SSNA controls cutaneous vasoconstriction

and sudomotor (sweating) activity. It is not pulse-synchronous but is irregular and bursty, responding to thermal needs and emotional arousal. The studies conclusively showed that **SSNA is markedly enhanced by local vibration**.¹

4.2 The 60 Hz "Danger Signal"

The specificity of the SSNA response is striking. When vibration frequencies were swept, the frequency of **60 Hz** was identified as the most effective stimulus for eliciting an increase in SSNA.¹

- **Mechanism:** This 60 Hz tuning suggests that the central nervous system (CNS) interprets afferent traffic from the upper range of the Meissner system (or the FA-I/FA-II crossover region) as a specific stress signal requiring a cutaneous defense response.
- **Acceleration Dependence:** At the fixed frequency of 60 Hz, the magnitude of the SSNA burst is linearly dependent on the vibratory acceleration.¹ This establishes a clear dose-response relationship: higher acceleration at 60 Hz yields stronger sympathetic outflow.
- **The Reflex Arc:** The proposed pathway is a "somato-sympathetic reflex."
 1. **Sensor:** Meissner/Pacinian corpuscles in the hand detect 60 Hz vibration.
 2. **Afferent Limb:** A-beta fibers transmit signals to the dorsal horn of the spinal cord and up to the brainstem.
 3. **Central Processing:** The signal is integrated in the hypothalamus or brainstem reticular formation, distinct from the vasomotor center controlling MSNA.
 4. **Efferent Limb:** Sympathetic C-fibers discharge selectively to the skin (SSNA), releasing norepinephrine and neuropeptide Y.
 5. **Effector:** Cutaneous arterioles constrict, and eccrine sweat glands activate (cold sweat).

4.3 Synergy with Environmental Noise

Industrial environments rarely present vibration in isolation; high noise levels are ubiquitous. The Nagoya findings indicate a powerful synergy: SSNA is "markedly enhanced" by the **combined** stimuli of local vibration and noise.¹ While noise alone triggers a startle-mediated sympathetic response, and vibration triggers the somato-sympathetic reflex, their combination produces a supralinear increase in vasoconstrictor tone. This suggests that the auditory and somatosensory inputs converge on the same autonomic nuclei, creating a state of hyper-arousal that accelerates vascular damage.¹

5. Vascular Pathophysiology: From Functional Spasm to Structural Remodeling

The progression from physiological reflex to pathological disease defines the natural history of VWF. The Nagoya studies provide rare and critical histological evidence linking the transient

sympathetic bursts described above to permanent structural changes in the blood vessels.

5.1 Acute Hemodynamics and Shear Stress

During acute vibration exposure, the surge in SSNA causes intense vasoconstriction of the digital arteries and arterioles. This leads to:

- **Flow Cessation:** Blood flow can drop to near zero, a state termed "vasospastic attack" or "blanching."
- **Shear Stress Alteration:** Vascular endothelial cells are sensitive to wall shear stress (WSS). Under normal flow, laminar shear stress promotes the release of vasodilators like Nitric Oxide (NO) and inhibits proliferation. The cessation of flow and the oscillatory nature of vibration-induced shear stress disrupt this homeostasis. Low or oscillatory WSS is a potent stimulus for pro-inflammatory signaling and smooth muscle proliferation.¹⁹
- **Endothelial Dysfunction:** Markers of endothelial damage, such as von Willebrand factor (vWF) and thrombomodulin, are elevated in the plasma of vibration-exposed workers.²¹ Additionally, there is a documented deficiency in the release of Calcitonin Gene-Related Peptide (CGRP), a potent neuropeptide vasodilator released by perivascular nerves.²¹ The loss of this "antagonist" to constriction leaves the vessel vulnerable to unchecked sympathetic drive.

5.2 Chronic Histopathology: Medial Hypertrophy and Fibrosis

Perhaps the most definitive finding from the Nagoya Journal (Vol 57) is the histological analysis of digital biopsies from vibration syndrome patients.²² Compared to age-matched controls, the vessels of VWF patients exhibit:

1. **Medial Hypertrophy:** The tunica media (muscle layer) of the small arteries and arterioles is significantly thickened. The smooth muscle cells have undergone hypertrophy (increase in size) and hyperplasia (increase in number).
2. **Perivascular Fibrosis:** There is a marked deposition of collagen fibers in the connective tissue immediately surrounding the vessels (the adventitia and perivascular space). This fibrosis acts as a rigid "cuff," preventing the vessel from expanding and likely compressing the vasa vasorum.
3. **Increased Wall-to-Lumen Ratio:** The ratio of the media thickness to the external diameter is significantly increased.²² This geometric remodeling encroaches on the lumen, increasing vascular resistance even in the absence of active tone. This explains why patients experience reduced blood flow even between attacks.

5.3 The "White Foot": Evidence for Systemic Dysregulation

A crucial controversy in HAVS research has been whether the damage is local (direct trauma to hand vessels) or systemic. The Nagoya study provided the "smoking gun" for the systemic theory by examining the **toes** of workers who operated hand-held tools (chainsaws) but had no direct vibration exposure to the feet.

- **Findings:** The digital arteries of the toes showed the **same** pathological changes—medial thickening and perivascular fibrosis—as the fingers.²²
- **Correlation:** The severity of the arterial thickening in the toes correlated significantly with that in the fingers of the same individual.²²
- **Implication:** Since the feet were never vibrated, the pathology cannot be due to local mechanical trauma. It must be the result of **chronic, repeated, centrally-mediated vasoconstriction**. The 60 Hz vibration in the hands triggers the central SSNA reflex, which discharges globally to all cutaneous beds, including the feet. Over years of exposure, this repeated sympathetic "clamping" drives structural remodeling in the toes just as it does in the hands.²³

6. Clinical Manifestations and Diagnostic Assessment

The physiological and pathological changes culminate in the clinical syndrome known as HAVS. The diagnosis relies on demonstrating the specific vascular and neurological deficits predicted by the underlying science.

6.1 Clinical Staging and Symptoms

The Stockholm Workshop Scale is the standard for grading HAVS.

- **Vascular Component:** Ranges from Stage 0 (no attacks) to Stage 4 (trophic changes/gangrene). The hallmark is the blanching attack, precipitated by cold or emotion, lasting minutes to hours.
- **Sensorineural Component:** Ranges from Stage 0SN (exposed, no symptoms) to Stage 3SN (severe numbness, loss of dexterity). This reflects the damage to the Pacinian/Meissner corpuscles and the demyelination of A-beta fibers.²⁵

6.2 The Cold Provocation Test (CPT)

To objectively assess the vascular hyper-reactivity, the Cold Provocation Test is employed.

- **Protocol:** Hands are immersed in water (typically 10°C or 12°C) for 5-10 minutes.²⁶
- **Physiological Basis:** Cold is a potent stimulus for SSNA (thermoregulatory vasoconstriction). In a healthy person, cold causes constriction, but blood pressure in the finger is maintained, and rewarming is rapid due to Cold-Induced Vasodilation (CIVD or the "hunting reaction").
- **Findings in VWF:**
 - **FSBP% (Finger Systolic Blood Pressure):** The ratio of systolic pressure in the cooled finger to the systemic pressure drops significantly in VWF patients (often <60%, compared to >80-90% in normals).²⁸ This indicates that the critical closing pressure of the hypertrophied vessels is easily reached under sympathetic load.
 - **Rewarming Time:** After removing the hand from cold water, VWF patients show a prolonged delay in the return of skin temperature (FST). While FSBP (arterial

pressure) may improve after years of ceasing exposure, the **thermal recovery often remains impaired**.²⁸ This dissociation suggests that while the active vasospasm may lessen, the downstream microcirculatory damage or the perivascular fibrosis (which limits vasodilation) is permanent.

- **Specificity:** The combination of FSBP% and rewarming curves provides a sensitivity of 84-99% and specificity of 94-100% for diagnosing VWF.²⁹

Table 2: Diagnostic Metrics in Vibration Syndrome

Metric	Healthy Subject	VWF Patient	Physiological Basis
FSBP% (at 10°C)	> 85 - 90%	< 60%	Critical closing pressure of hypertrophied arteries.
Rewarming Time	Rapid (< 5-10 min)	Delayed (> 15-20 min)	Impaired CIVD and endothelial vasodilator function.
Vibrotactile Threshold	Low	Elevated	Degeneration of Pacinian/Meissner corpuscles.
SSNA Response	Moderate	Hyper-reactive	Sensitization of the somato-sympathetic reflex.

7. Regulatory Frameworks and the Frequency Weighting Debate

The translation of this biological knowledge into safety standards has been fraught with controversy. The current international standard, **ISO 5349**, uses a frequency-weighting curve (\$W_h\$) to estimate the risk of injury. A critical analysis of the Nagoya findings and follow-up literature reveals significant discrepancies between this standard and physiological reality.

7.1 The ISO 5349 Weighting Curve (\$W_h\$)

The ISO 5349 standard assumes that the damaging potential of vibration is inversely proportional to frequency. The W_h curve is a low-pass filter: it gives maximum weight to frequencies between 8 Hz and 16 Hz and progressively attenuates the weight of higher frequencies (a slope of -6 dB/octave).

- **The Rationale:** This weighting was originally derived from studies on subjective discomfort and equal sensation contours, not pathology.⁹ It assumes that displacement is the primary damage vector, and since displacement decreases with frequency (for constant acceleration), risk decreases.

7.2 The Biological Discrepancy (The "Greenstein/Bovenzi" Critique)

The accumulated evidence strongly suggests that the ISO 5349 weighting **underestimates** the risk posed by high-frequency vibration (>50 Hz).⁴

1. **Vascular Risk:** Epidemiological studies of stone workers and foresters show that high-frequency tools (like grinders and chainsaws) cause VWF at exposure levels that ISO 5349 predicts should be safe. Conversely, low-frequency high-magnitude vibration (like compactors) causes less VWF than predicted.⁵
2. **Physiological Misalignment:**
 - **Reflex Trigger:** The Nagoya studies pinpoint **60 Hz** as the peak SSNA trigger.¹ The ISO curve begins to discount risk significantly at this frequency.
 - **Resonance:** Finger resonance occurs between **100 Hz and 300 Hz**.⁸ At these frequencies, energy absorption in the digits is maximal. The ISO curve heavily discounts this band.
 - **Receptor Damage:** The Pacinian corpuscle is most sensitive at **250 Hz**. To prevent sensorineural damage, the standard should weight this frequency band heavily. ISO 5349 does the opposite.
3. **Alternative Weightings:** Researchers like Bovenzi and Tominaga have proposed alternative weightings (W_p or W_{hf}) that maintain a flat response or higher weighting up to 200–400 Hz.⁵ These curves provide a much better statistical fit to the prevalence of VWF in worker populations.

7.3 Implication for Prevention

The reliance on ISO 5349 leads to the design of "anti-vibration" gloves and tools that dampen low frequencies (which are easier to isolate) but may fail to attenuate—or even amplify—high frequencies in the 100–300 Hz range. If the biological driver of the SSNA reflex (60 Hz) and the direct tissue damager (>100 Hz) are not adequately suppressed, the protective equipment provides a false sense of security while the pathogenesis of VWF proceeds unchecked.

8. Synthesis and Future Directions

The integration of the *Nagoya Journal of Medical Science* findings with modern biophysics

and molecular biology constructs a coherent narrative of Hand-Arm Vibration Syndrome as a systemic autonomic pathology.

The causal chain is clear:

1. **Exposure:** Industrial vibration containing significant energy in the **60 Hz - 300 Hz** band.
2. **Transduction:**
 - **60 Hz:** Resonates with the hand-arm system and optimally drives **Meissner's corpuscles**, triggering a central **somato-sympathetic reflex**.
 - **>100 Hz:** Resonates with digital tissues, driving **Pacinian corpuscles** and causing direct shear damage to the **endothelium**.
3. **Autonomic Response:** A specific, profound increase in **SSNA** (but not MSNA), exacerbated by **noise**.
4. **Pathology:**
 - **Acute:** Vasospasm and shear stress reduction.
 - **Chronic:** Systemic **medial hypertrophy** and **fibrosis** of cutaneous arteries, affecting both hands and **feet**.
 - **Neural:** Piezo2 channel dysfunction and mechanoreceptor degeneration.

This synthesis mandates a paradigm shift in both clinical management and regulation. Clinically, the assessment of vibration injury must recognize the systemic nature of the disorder—checking the feet is as important as checking the hands. Regulators must urgently revise exposure standards to account for the high-frequency neurovascular toxicity that the current ISO 5349 weighting obscures. Future research should focus on the genetic basis of Piezo2 susceptibility and the development of pharmacological agents that can specifically uncouple the aberrant SSNA reflex without compromising homeostatic blood pressure control.

9. Addendum: Technical Data Summary

Table 3: Pathological Morphometrics of Digital Arteries (Nagoya Findings)

Source Data: Hashiguchi T, et al. ²²

Measurement	Control (Finger)	Patient (Finger)	Control (Toe)	Patient (Toe)	Statistical Significance
Media Thickness	Normal	Marked Hypertrop	Normal	Marked Hypertrop	p < 0.01

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Media/External Diameter Ratio	Low	High	Low	High	p < 0.01
Perivascular Collagen	Sparse	Dense/Fibrotic	Sparse	Dense/Fibrotic	Qualitative observation
Intimal Changes	Minimal	Variable	Minimal	Variable	Less consistent than medial changes

Table 4: Frequency Response of the Hand-Arm System

Source Data: ⁶

Component	Resonance Frequency (fn)	Biomechanical Behavior	Pathological Risk
Shoulder/Arm	< 10 Hz	High transmissibility	Musculoskeletal pain (neck/shoulder)
Hand-Arm Unit	16 - 62.5 Hz	Amplification at wrist	Wrist osteoarthritis, SSNA Reflex Trigger (60 Hz)
Individual Fingers	100 - 300 Hz	Local absorption	VWF, Digital Neuropathy, Endothelial Shear
Skin/Pulp	> 300 Hz	Surface wave propagation	Meissner/Pacinian desensitization

Table 5: SSNA vs. MSNA Response Specificity

Source Data: Mano T. ¹

Stimulus	SSNA Response	MSNA Response	Mechanism
Vibration (60 Hz)	+++ (Marked Increase)	0 (No Change)	Somato-sympathetic reflex (Cutaneous specific)
Noise	+ (Startle)	0 / + (Variable)	Auditory-sympathetic reflex
Vibration + Noise	++++ (Synergistic)	0 / +	Cross-modal integration / Hyper-arousal
Static Exercise	+	+++	Metaboreflex (Muscle specific)
Gravity (+Gz)	0	+++	Baroreflex (Hemodynamic defense)

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